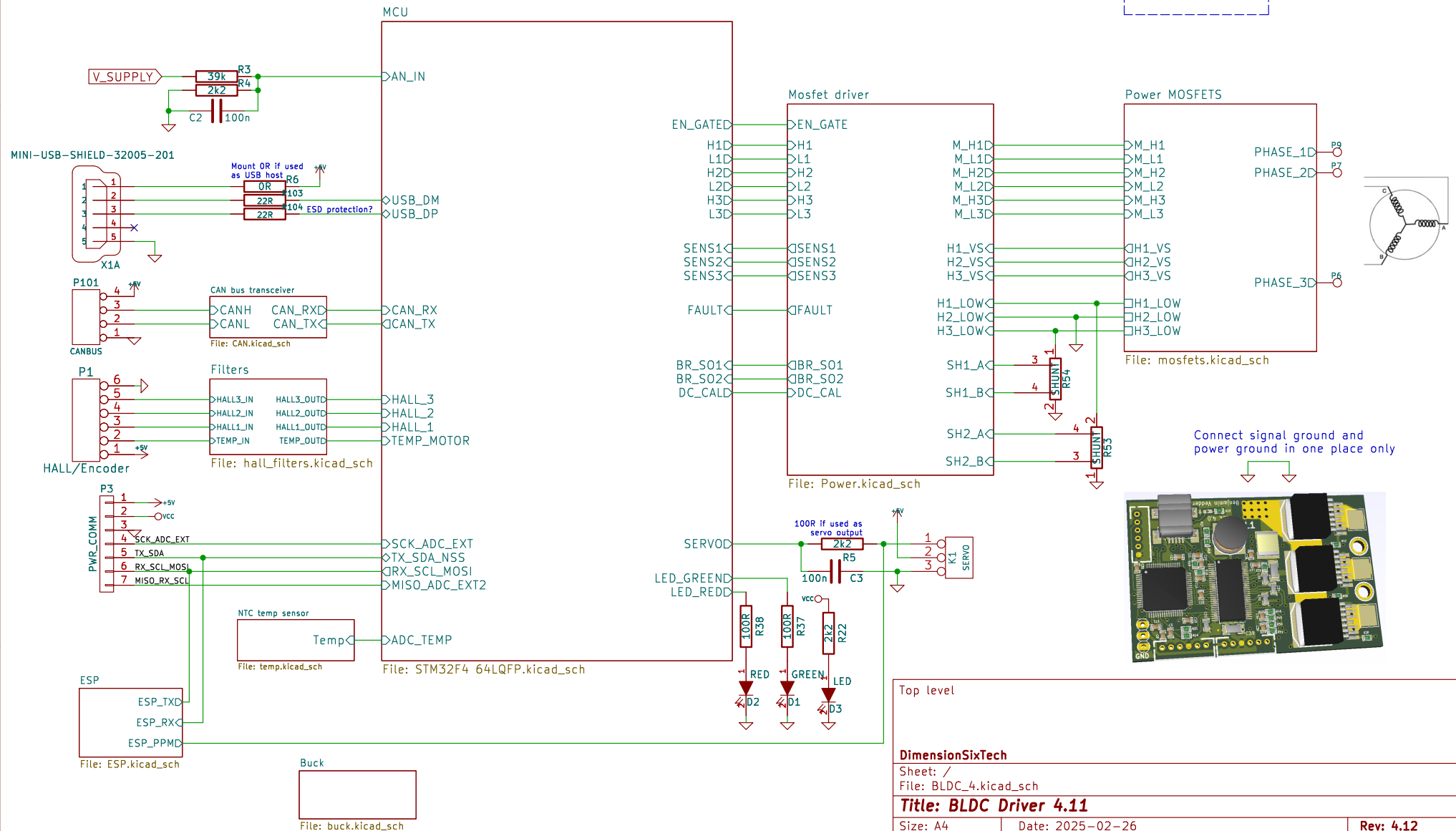


BLDC motor controller



Top level

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Sheet: /
File: BLDC_4.kicad_sch

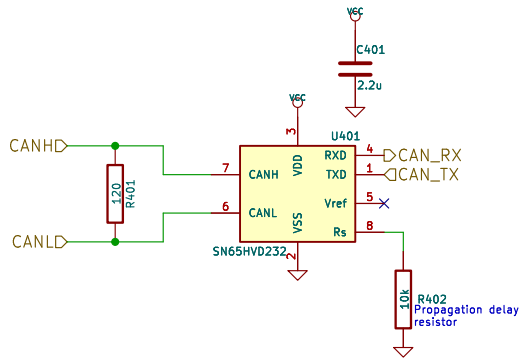
Title: BLDC Driver 4.11

Size: A4 Date: 2025-02-26

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Package Types

MCP2561
PDIP, SOIC

TxD	1	8	STBY
VSS	2	7	CANH
VDD	3	6	CANL
RxD	4	5	SPLIT

MCP2562
PDIP, SOIC

TxD	1	8	STBY
VSS	2	7	CANH
VDD	3	6	CANL
RxD	4	5	Vio

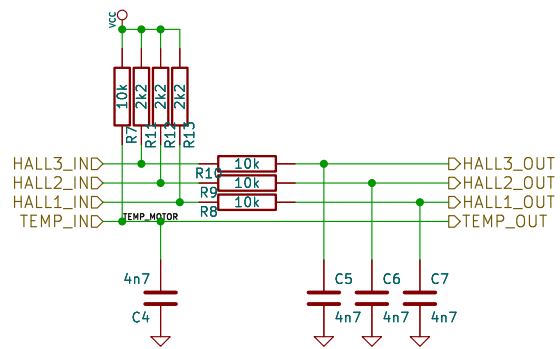
MCP2561
3x3 DFN*

TxD	1	O	8	STBY
VSS	2	EP	7	CANH
VDD	3	9	6	CANL
RxD	4		5	SPLIT

MCP2562
3x3 DFN*

TxD	1	O	8	STBY
VSS	2	EP	7	CANH
VDD	3	9	6	CANL
RxD	4		5	Vio

* Includes Exposed Thermal Pad (EP); see [Table 1-2](#).



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Title: BLDC Driver 4.11

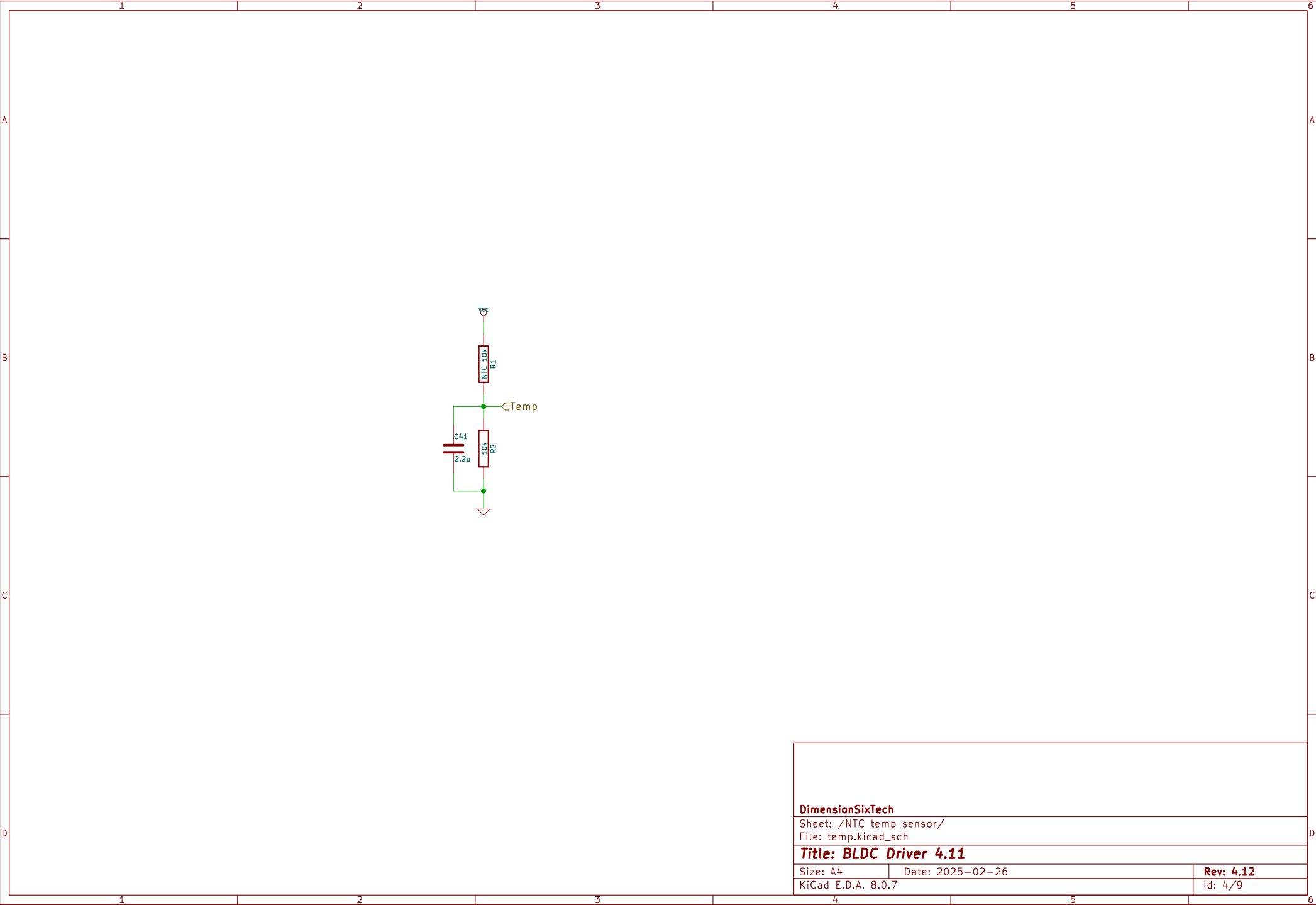
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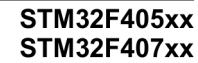
Date: 2025-02-26

Rev: 4.12

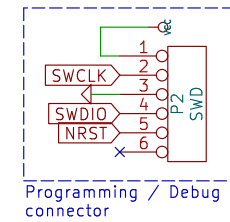
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Id: 3/9



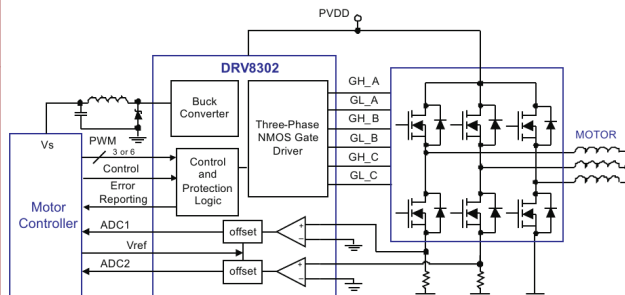
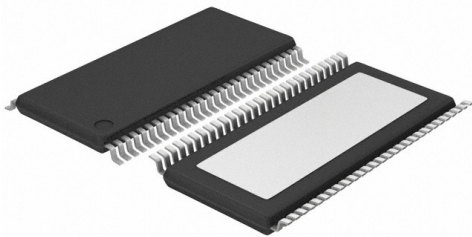


A circuit diagram showing five 2.2uF capacitors (C10, C11, C12, C13, C14) connected in parallel between VCC and GND. A note at the bottom says "Bypass, place next to VDD pins".

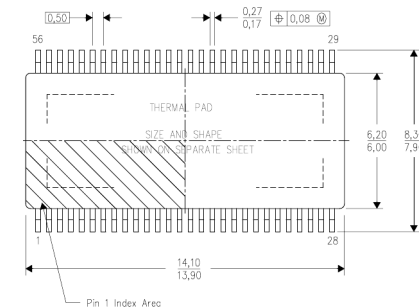
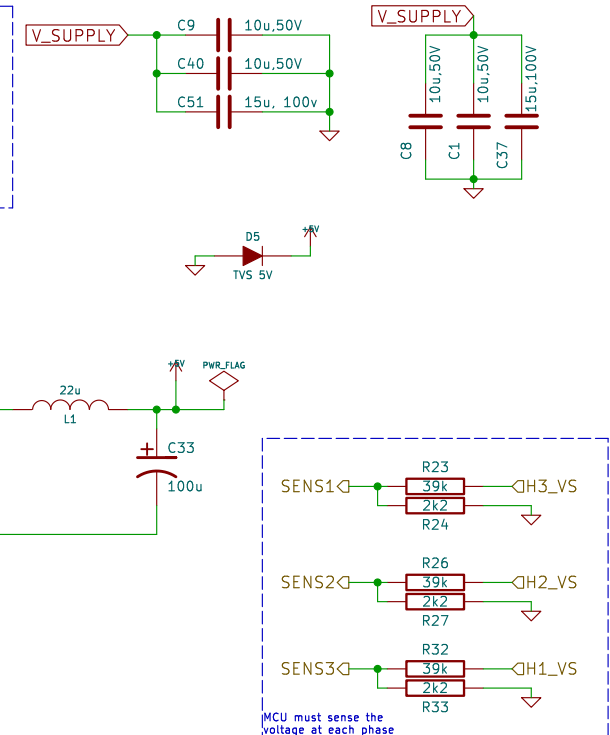


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- Operating Supply Voltage 8V–60V
- 2.3A Sink and 1.7A Source Gate Drive Current Capability
- Integrated Dual Shunt Current Amplifiers With Adjustable Gain and Offset
- Integrated Buck Converter to Support up to 1.5A External Load
- Independent Control of 3 or 6 PWM Inputs
- Bootstrap Gate Driver With 100% Duty Cycle Support
- Programmable Dead Time to Protect External FETs from Shoot Through
- Programmable Overcurrent Protection of External MOSFETs
- Thermally Enhanced 56-Pin TSSOP Pad Down DCA Package



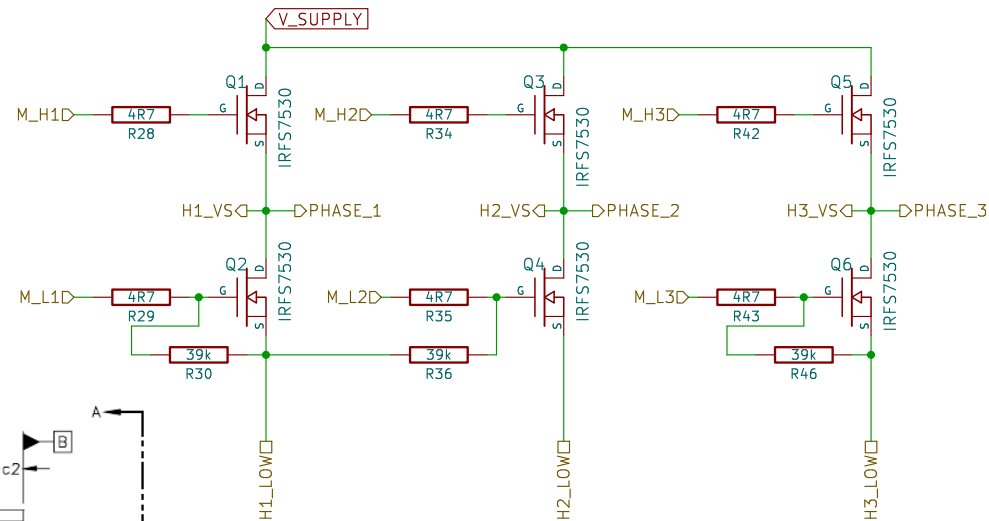
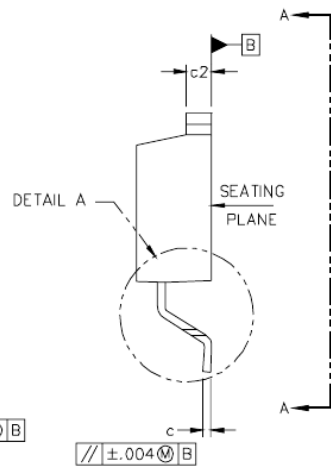
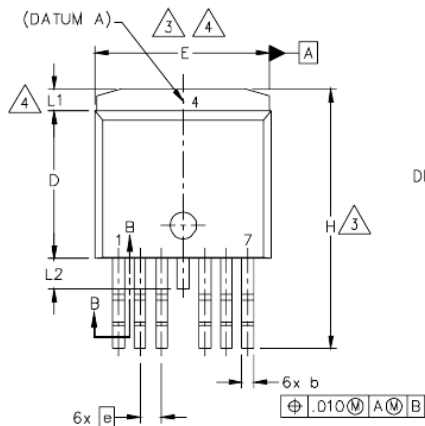
RECOMMENDED OPERATING CONDITIONS			MIN	TYP	MAX	UNITS
PVDD01	DC supply voltage PVDD01 for normal operation	Relative to PFGND	8	60	V	
PVDD02	DC supply voltage PVDD02 for bank conversion		3.5	60	V	
GVDD0	External capacitation on AVDD pin (operating at 20% tolerance)		1		μF	
GVDD01	External capacitation on PVDD01 pin (operating at 20% tolerance)		1		μF	
GVDD02	External capacitation on PVDD02 pin (operating at 20% tolerance)		2.2		μF	
GVDD03	Flying cap on charge pump pins (between CP1 and CP2) (operating cap 20% tolerance)		22		nF	
GVDD04	Bootstrapping cap (operatic cap)		22		nF	
ICL01	Input current of digital pin when EN_GATE is high			100	μA	
ICL02	Input current of digital pin when EN_GATE is low			1	μA	
ICL03	Maximum capacitation on digital input pin			100	pF	
TS_1pin	Maximum total input capacitance on each digital pin	20 pF				
R0101	Dead time control resistor range. Time range is 50ns (50ns) to 500ns (150ns) with 20pF of parasitic		0	150	kΩ	
RA01	FAULT pin sink current. Open drain	V = 4.4 V	2		mA	
RA02	OTW pin sink current. Open drain	V = 4.4 V	2		mA	
TA01	Temperature range of the current source		-40		125	°C
TA02	Operating switching frequency of gate driver	Gat1(G1) = 25 kHz or total 30 mA gate drive average current	200		kHz	
TA03	Ambient temperature		-40		125	°C



Sheet: /Mosfet driver/
File: Power.kicad_sch

Size: A4	Date: 2025-02-26
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Sheet: /Power MOSFETS/

File: mosfets.kicad_sch

Title: BLDC Driver 4.11

Size: A4

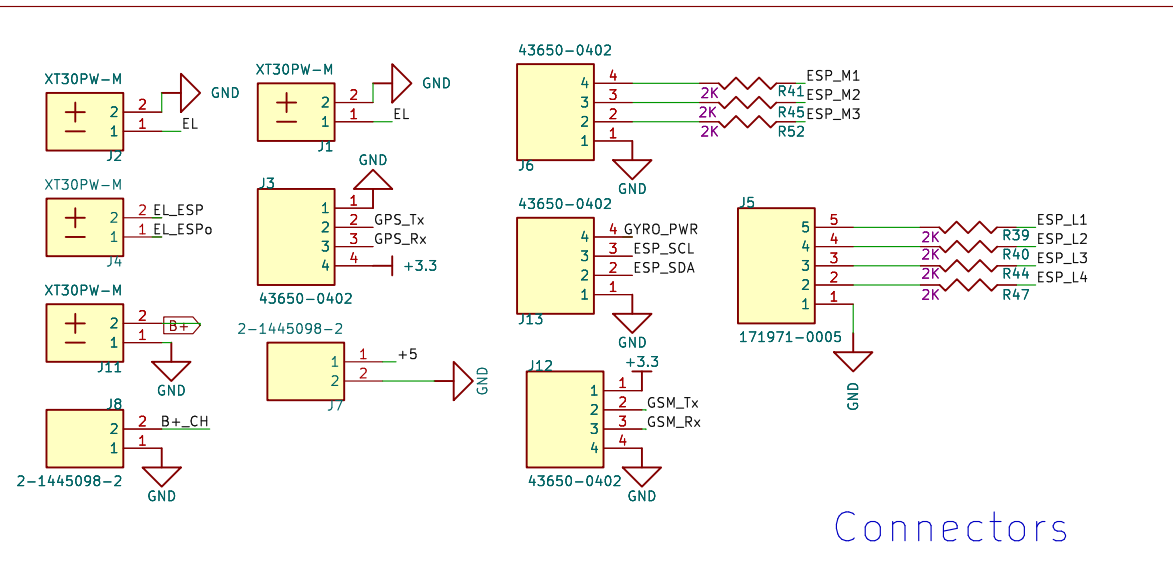
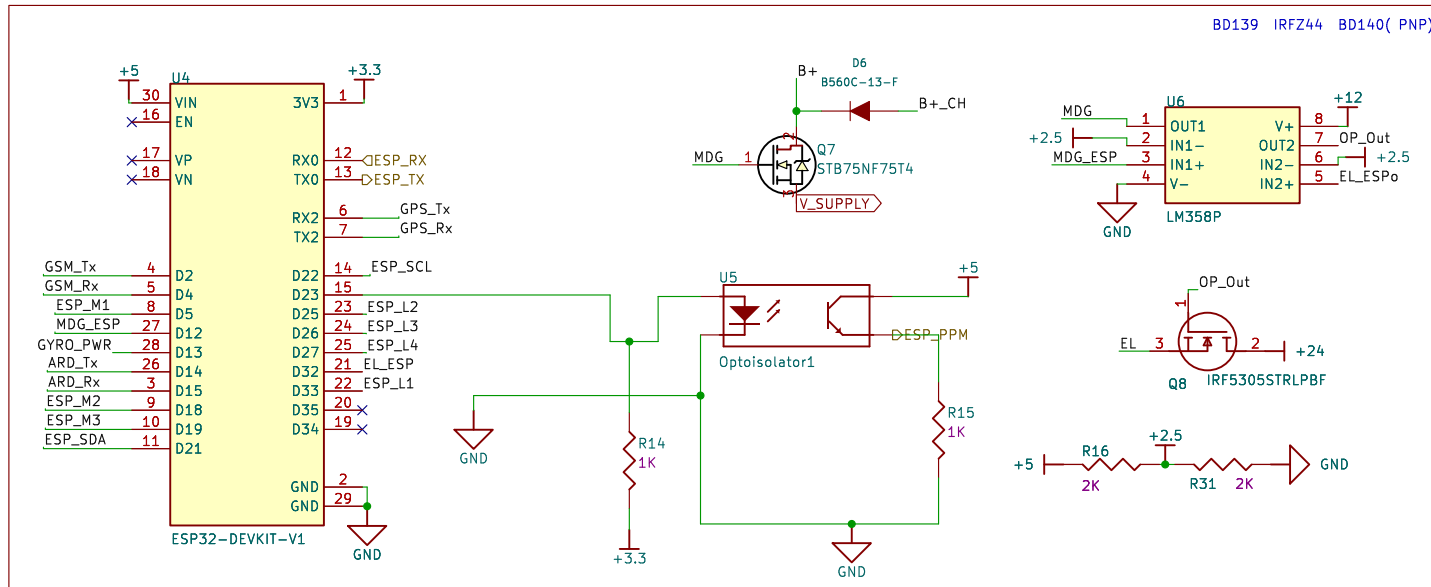
Date: 2025-02-26

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Current Design



Connectors

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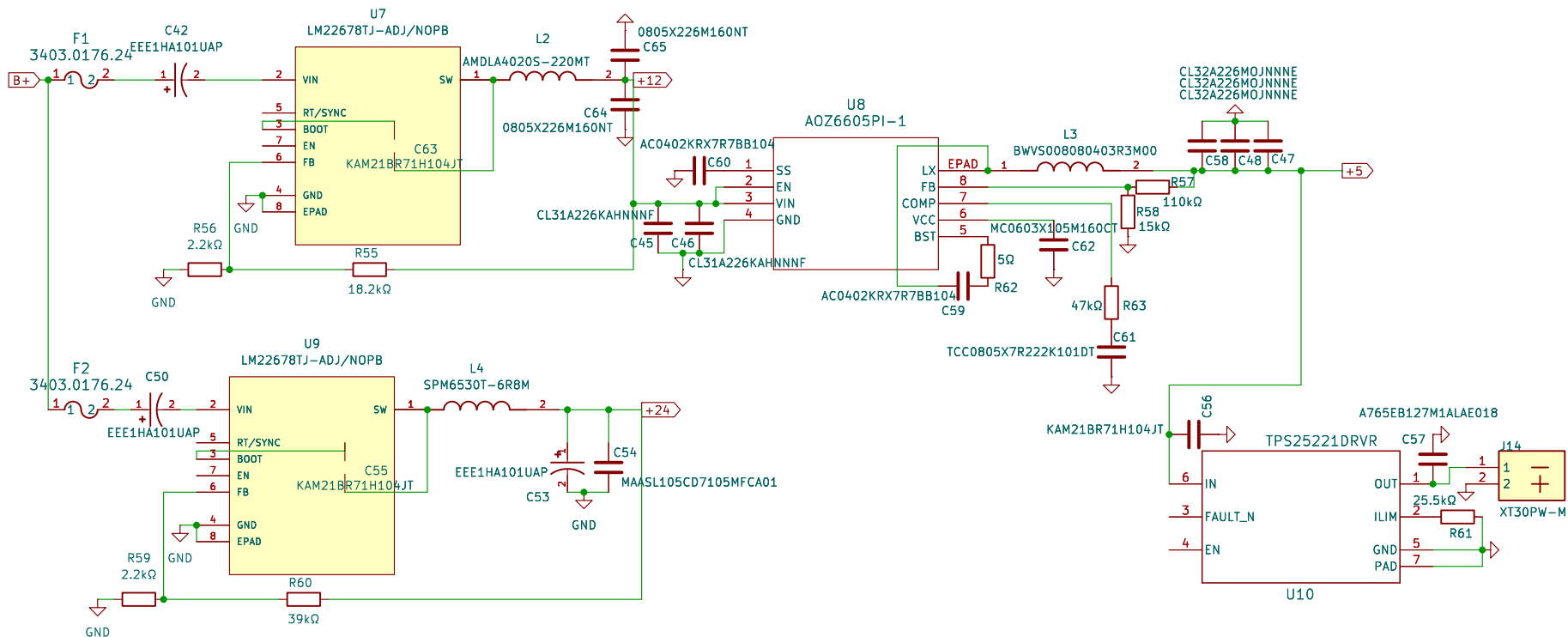
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Size: A4	Date: 2025-02-26
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Size: A4	
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Rev: 4.12

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DimensionSixTech

Sheet: /Buck/
File: buck.kicad_sch

Title: BLDC Driver 4.11

Size: A4 Date: 2025-02-26

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Rev: 4.12

Id: 9/9